

## **Implement SSL-2,3,4 (Task-5&6)**

### **Group-9**

#### **Group Members**

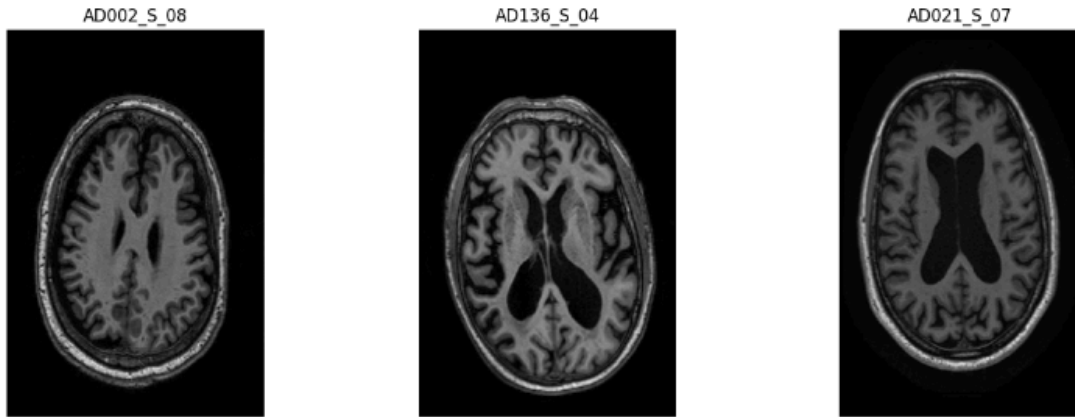
| <b>Name</b>         | <b>ID</b>     |
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| Miftha Ahona Majid  | 2022-1-50-015 |
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| Eshrat Jahan Eshi   | 2020-1-60-383 |

#### **Submitted To:**

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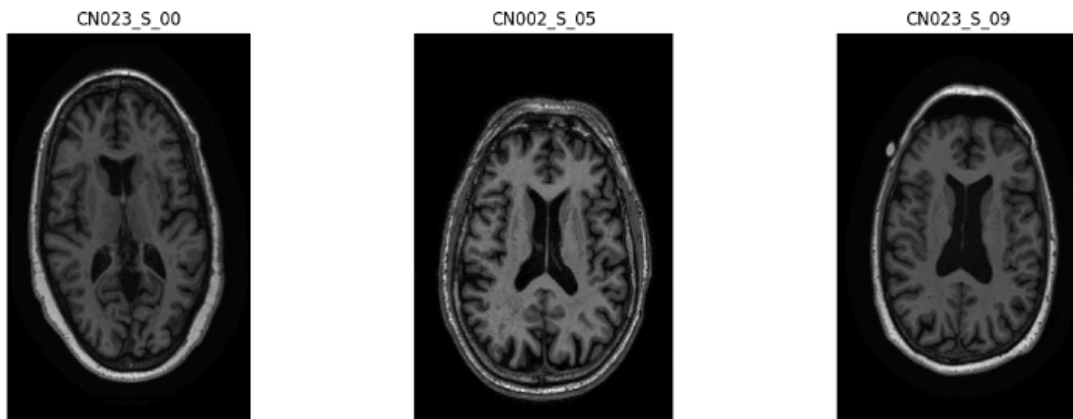
## Test Dataset Samples ( AD, CN, MCI)

### AD (Test)



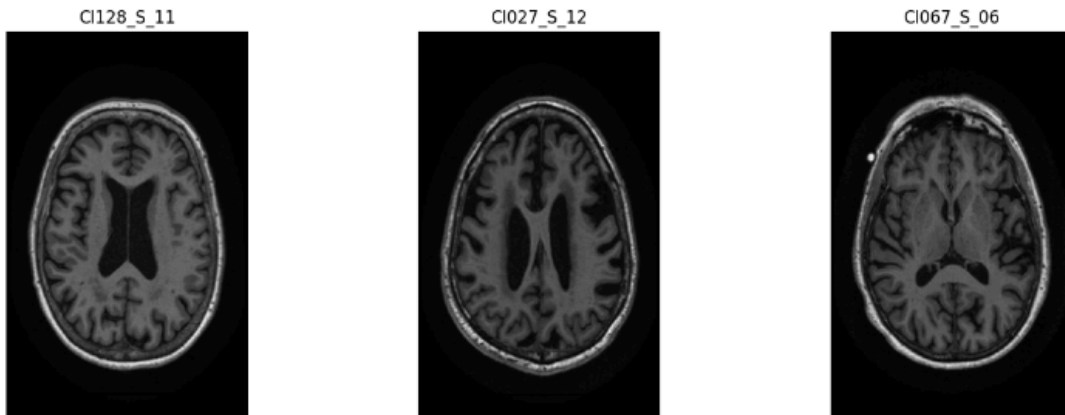
This image represents a brain scan from the test dataset labeled as **Alzheimer's Disease (AD)**. It typically shows structural abnormalities such as cortical thinning and enlarged ventricles. Used to evaluate model performance in identifying AD cases.

### CN (Test)



A test image labeled as **Cognitively Normal (CN)**. Displays a healthy brain structure with no visible signs of degeneration. Serves as a baseline for comparison with AD and MCI scans.

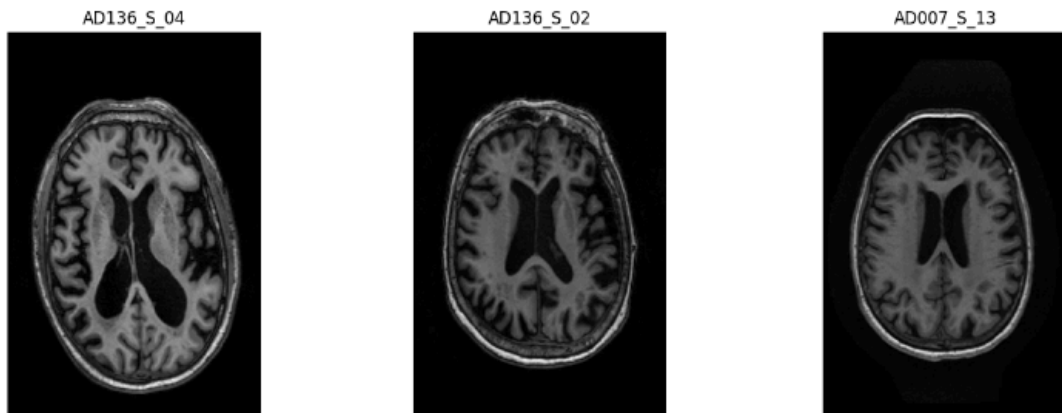
### MCI (Test)



This test image is labeled as **Mild Cognitive Impairment (MCI)**. Shows subtle changes in brain structure, intermediate between CN and AD. Helps assess the model's sensitivity to early-stage cognitive decline.

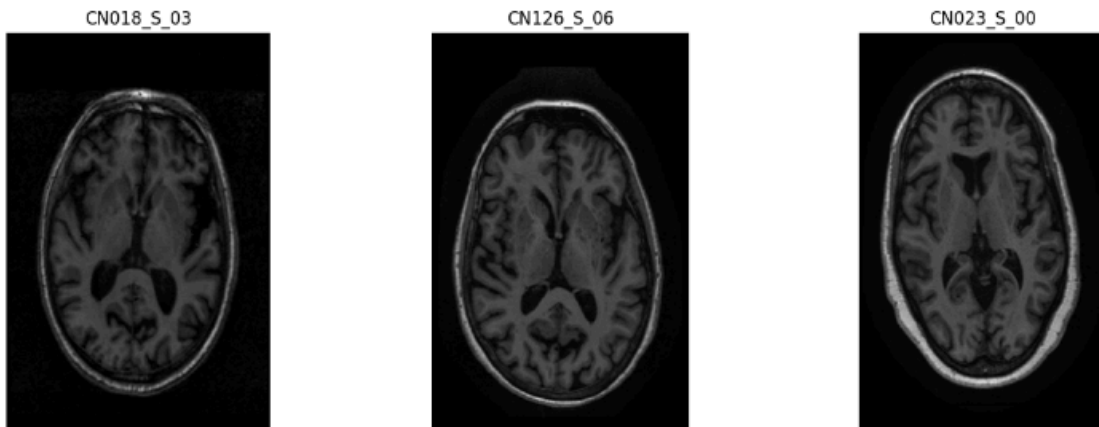
### Train Dataset Samples ( AD, CN, MCI)

#### AD (Train)



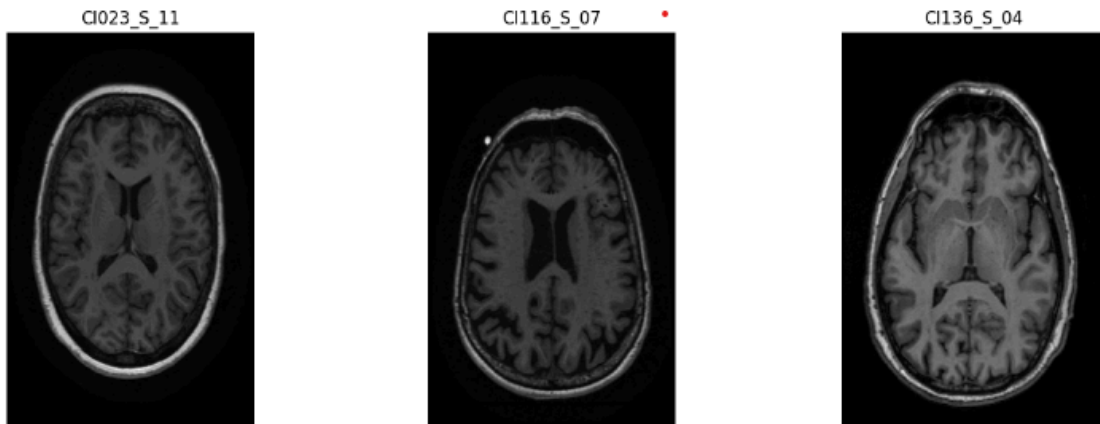
A training image labeled as **Alzheimer's Disease (AD)**. Used to teach the model to recognize patterns associated with AD. Features similar to the AD test image, aiding in model generalization.

#### CN (Train)



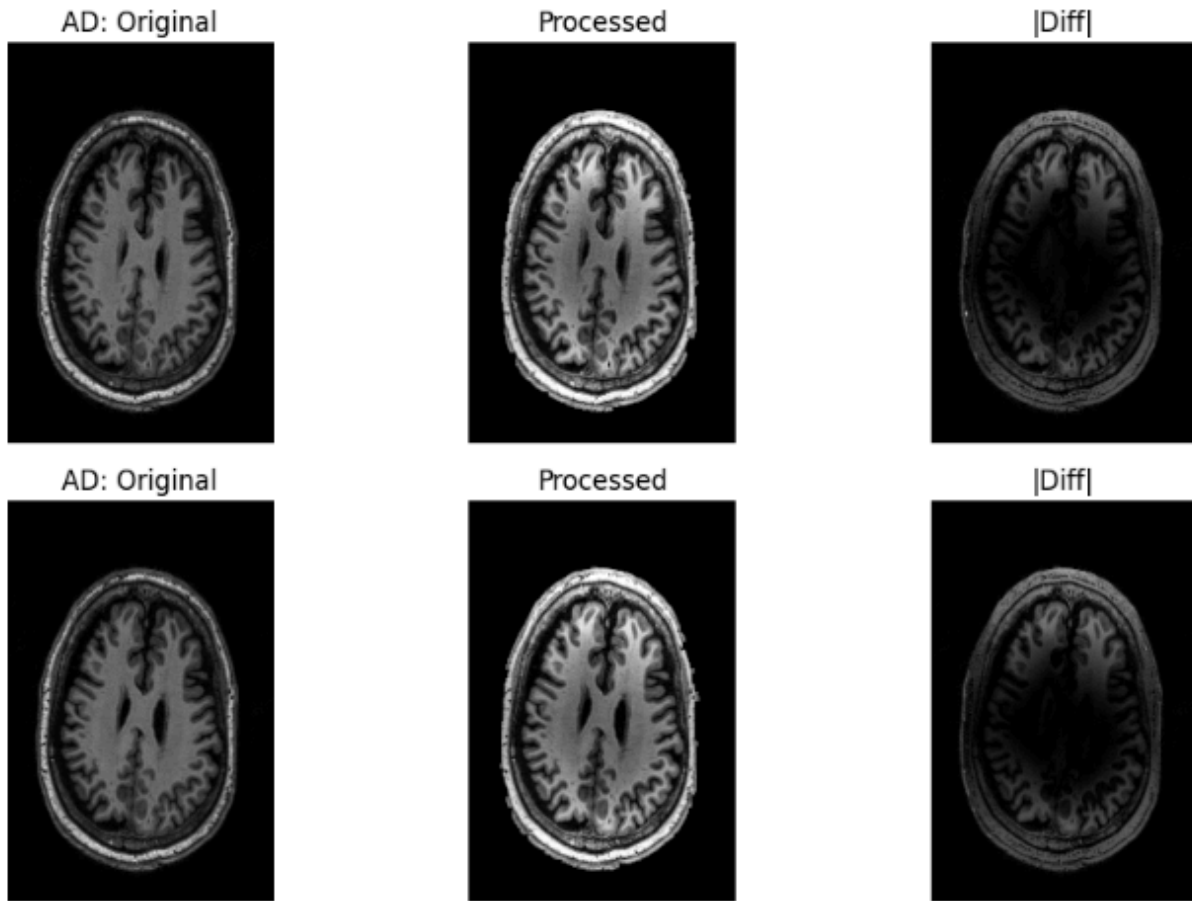
A training image labeled as **Cognitively Normal (CN)**. Represents normal brain anatomy, helping the model learn what healthy scans look like. Crucial for distinguishing CN from MCI and AD.

#### MCI (Train)



A training image labeled as **Mild Cognitive Impairment (MCI)**. Provides examples of early cognitive decline for model learning. Important for improving classification accuracy between CN and AD.

## AD Original- Processed- Difference

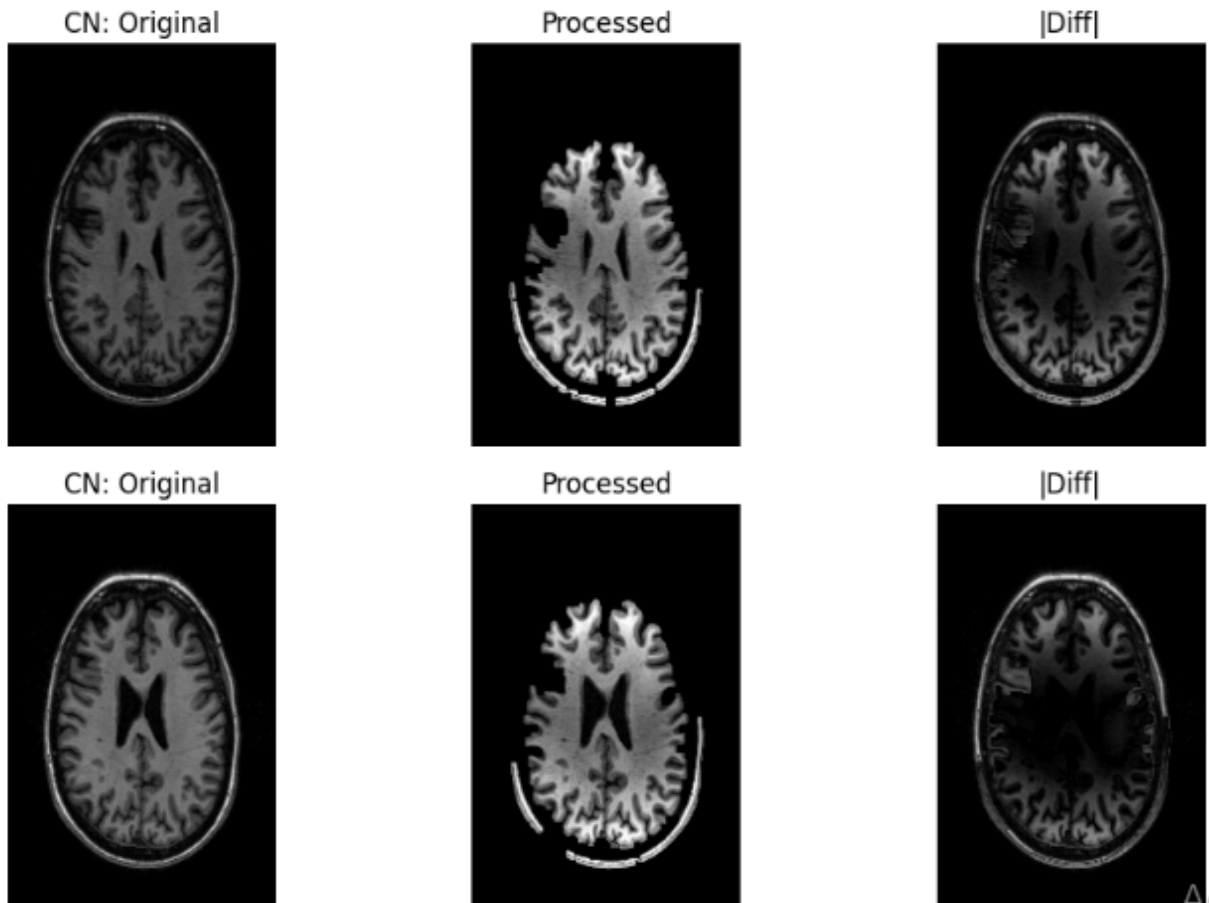


**Original:** Displays the raw brain scan of a patient diagnosed with Alzheimer's Disease.

**Processed:** Shows the image after preprocessing steps like normalization, resizing, or filtering.

**Difference:** Highlights the changes between the original and processed image, often emphasizing regions affected by AD such as hippocampal shrinkage or cortical thinning.

## CN Original- Processed- Difference

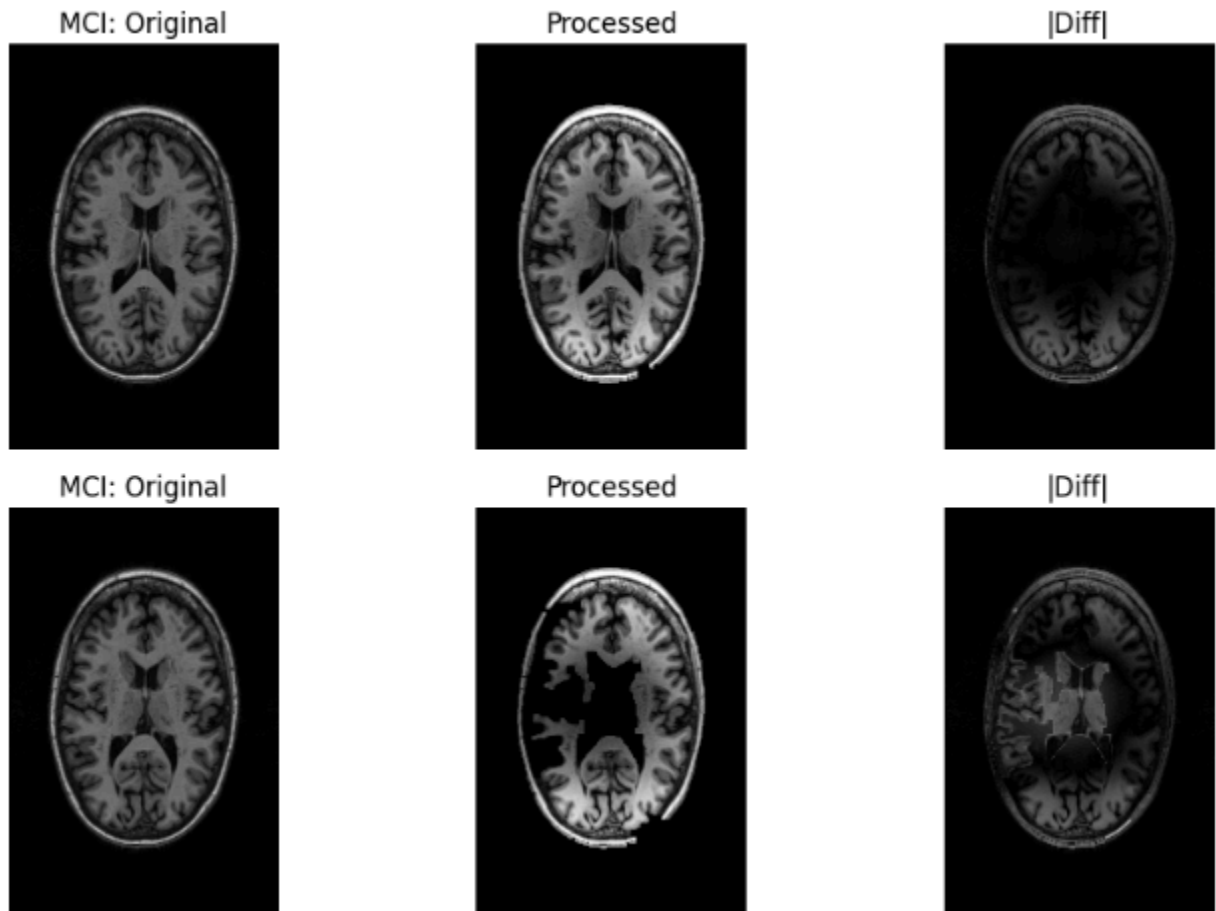


**Original:** A clean brain scan of a cognitively normal individual.

**Processed:** The image after enhancement and standardization for model input.

**Difference:** Minimal changes are visible, indicating stable brain structure with no signs of degeneration.

## MCI Original- Processed- Difference



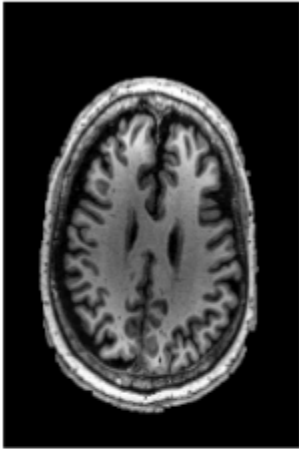
**Original:** Brain scan of a patient with Mild Cognitive Impairment.

**Processed:** Preprocessed version optimized for feature extraction.

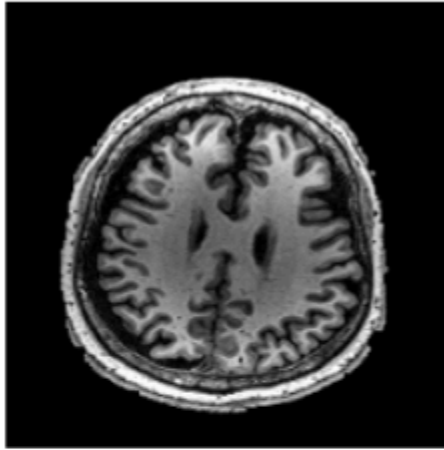
**Difference:** Shows subtle structural changes, such as mild atrophy, which are crucial for early detection and classification.

## Original (256x170) vs. Resized (224x224) images for AD, CN, and MC

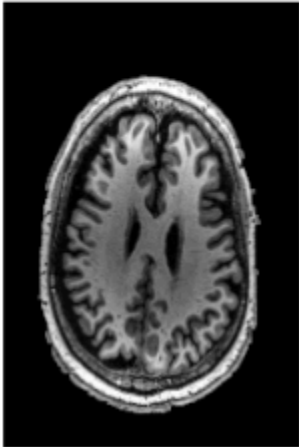
AD: Original (256, 170)



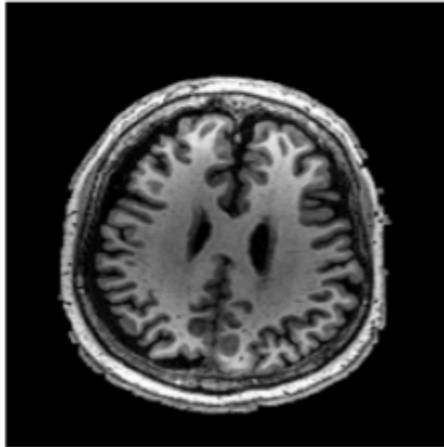
Resized (224, 224)



AD: Original (256, 170)

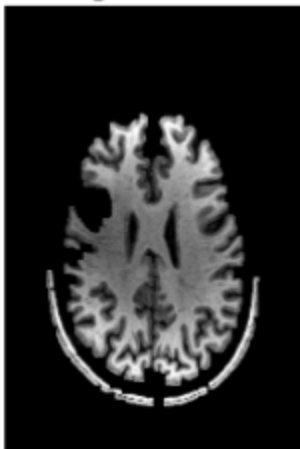


Resized (224, 224)

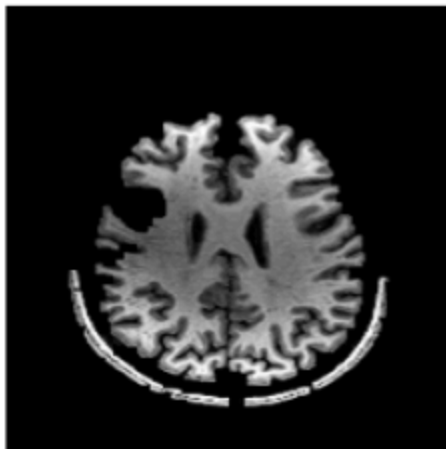




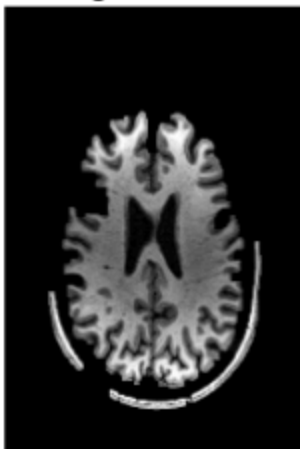
CN: Original (256, 170)



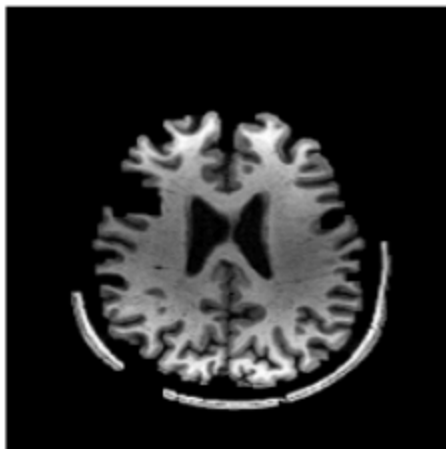
Resized (224, 224)



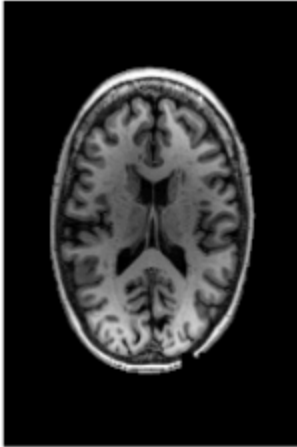
CN: Original (256, 170)



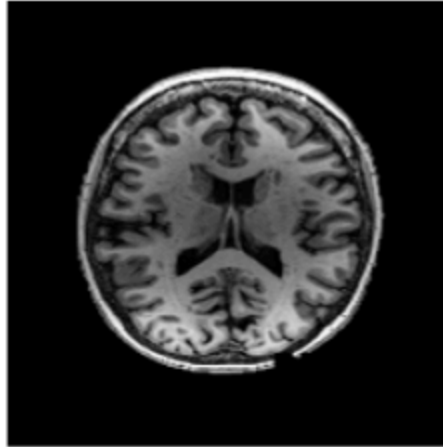
Resized (224, 224)



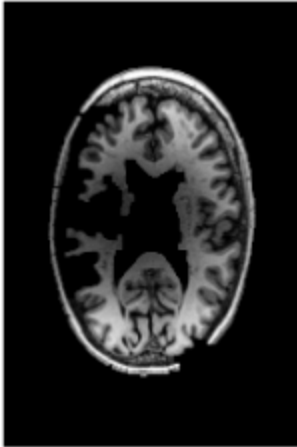
MCI: Original (256, 170)



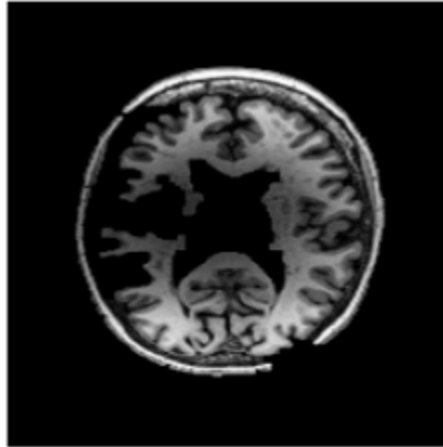
Resized (224, 224)



MCI: Original (256, 170)

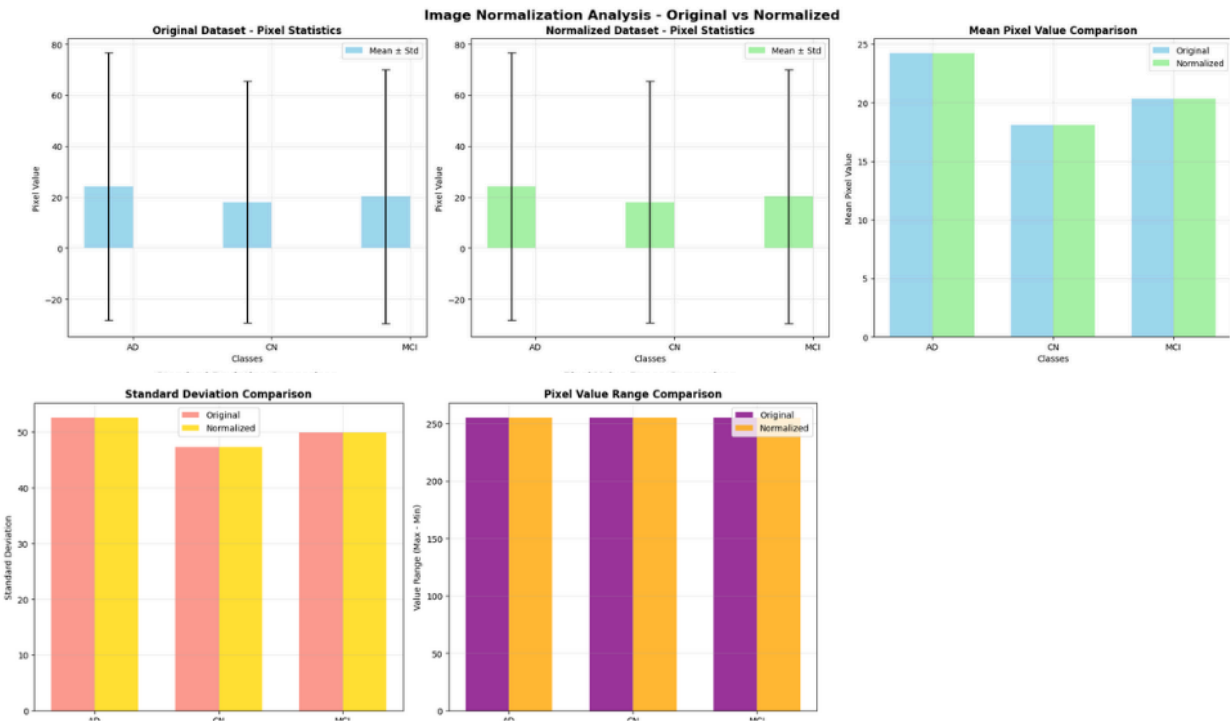


Resized (224, 224)

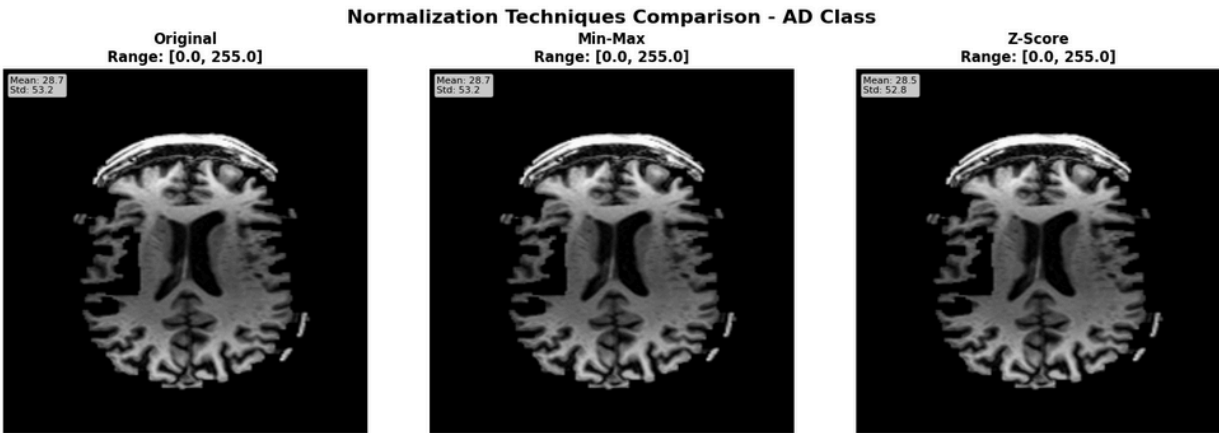


These images demonstrate standardization of input dimensions for deep learning models.

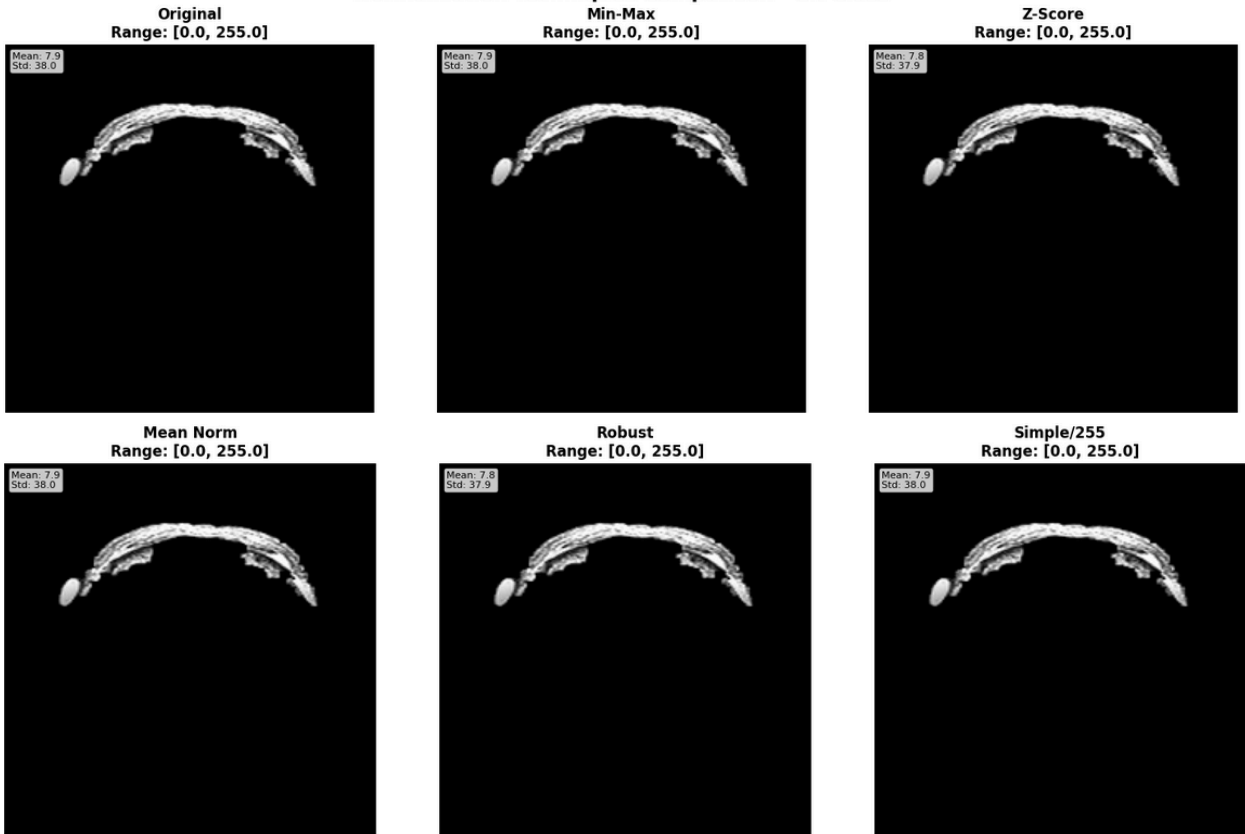
# Image Normalization Analysis



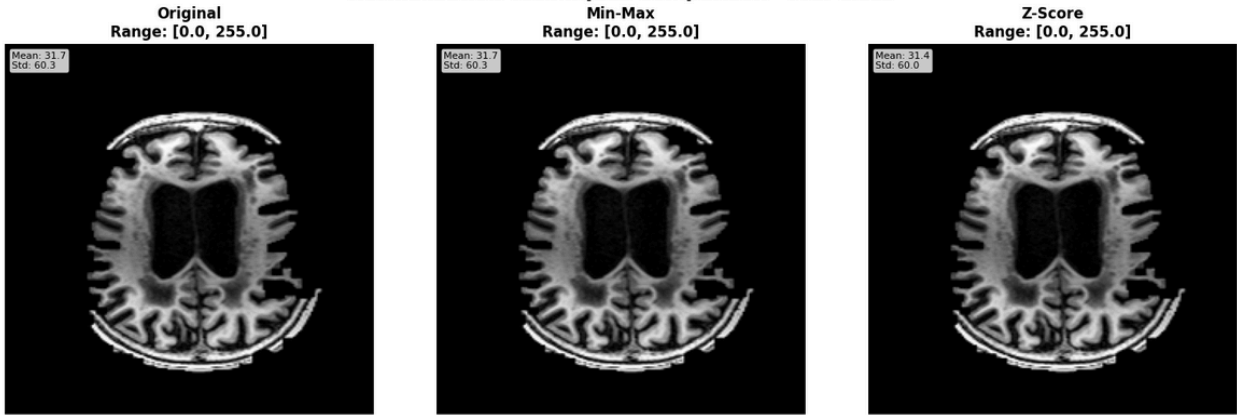
## Normalization Techniques Comparison

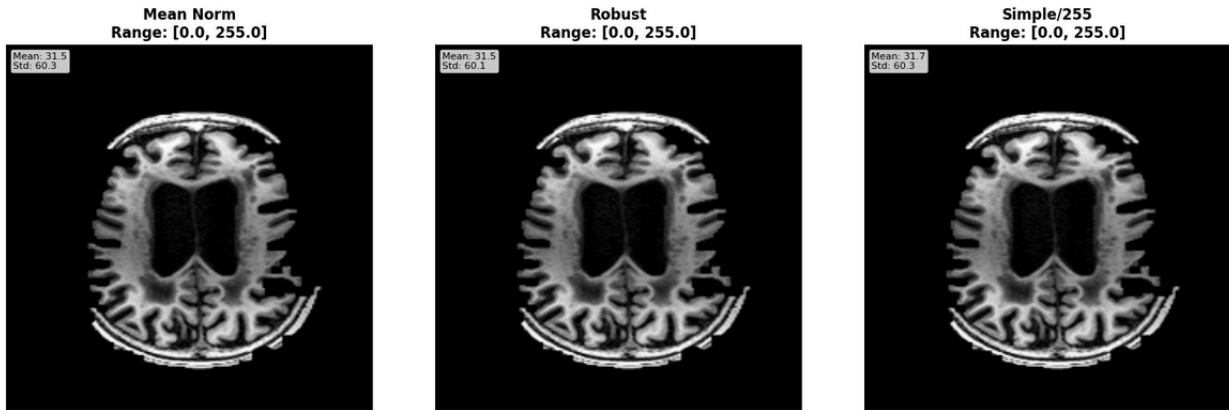


Normalization Techniques Comparison - CN Class



Normalization Techniques Comparison - MCI Class

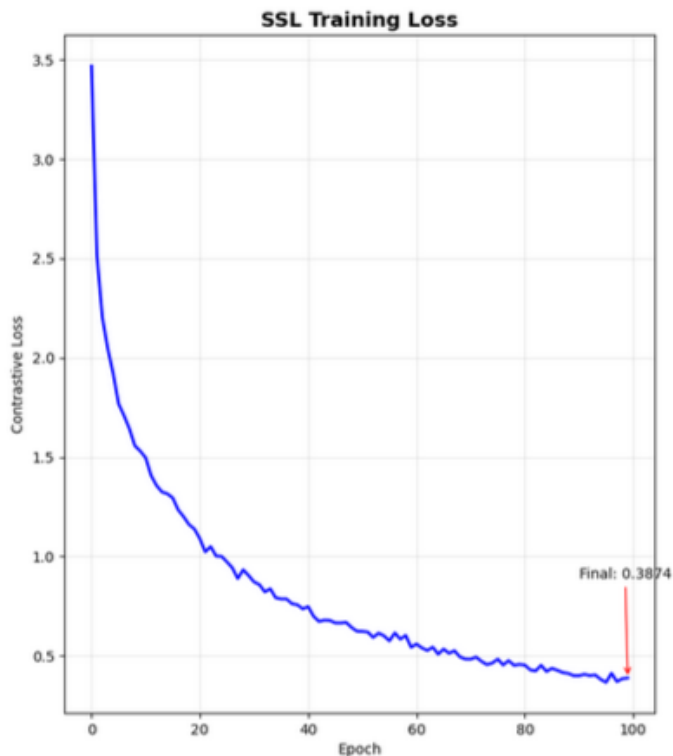




Those figures compare different normalization methods applied to AD,CN,MCI class images. While the visual appearance remains similar, each method changes the underlying pixel brightness distribution. Techniques like Z-score, Min-Max, Mean, and Robust normalization adjust brightness statistics differently, helping the model learn more stability and identifying features during training.

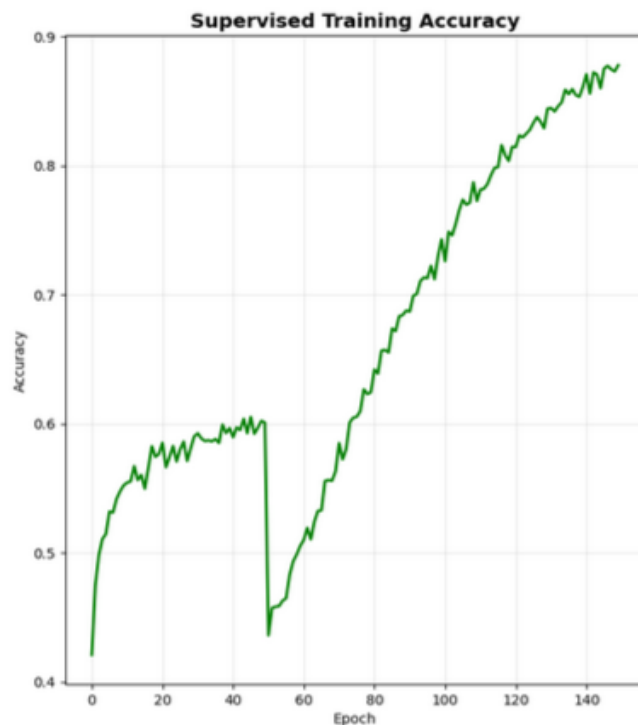
## SSL-2 SimCLR Method

### SSL Training Loss :



This graph shows a **SimCLR Training loss curve** labeled “Positive Start.” The y-axis represents **loss values** ranging from 0 to 3.5. The x-axis represents **epochs** from 0 to 100. This graph shows how the loss decreases during the self-supervised learning phase. Over the training epochs, the loss steadily goes down, indicating the model is learning meaningful representations. Near the **final epochs(100)**, the contrastive loss at around **0.3874**

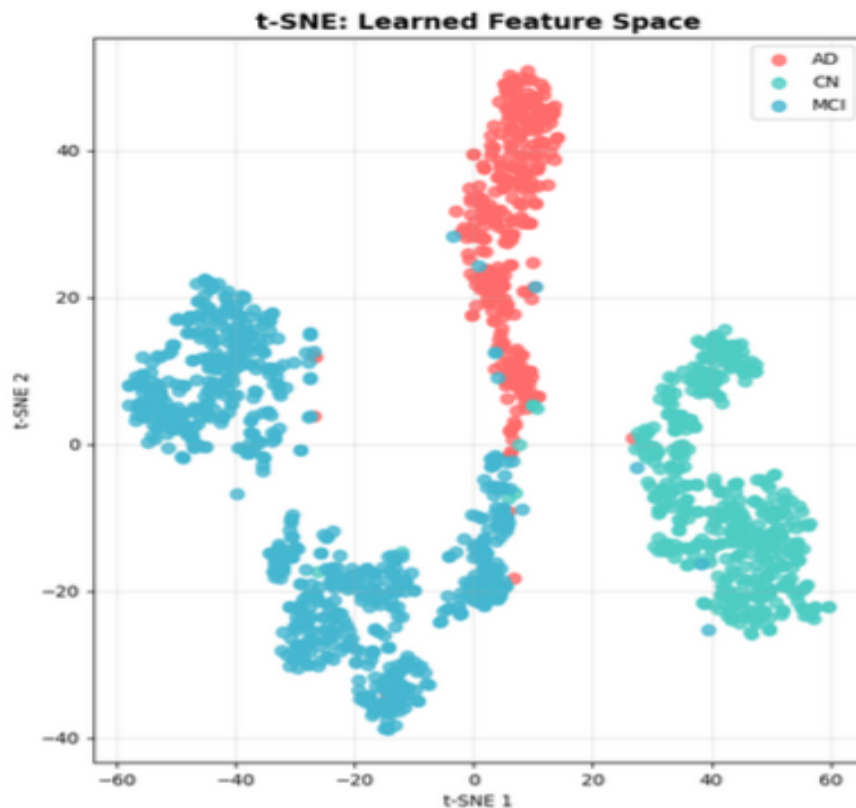
## Training Accuracy:



This graph displays the accuracy during supervised training. The y-axis represents **accuracy** ranging from 0.4 to 0.9. The x-axis represents **epochs** from 0 to 140. At the beginning, accuracy

increases rapidly, but **around 45 epochs** it **decreases sharply**. Over time the improvements become smaller but continue to go up, then reaching **around ~85–90%**, which indicates strong performance on the labeled dataset.

**t-SNE Learned Feature Space:**



This shows the t-SNE projection of the learned feature space. Each point represents a data sample embedded in 2D space. Different colors indicate different classes (AD, CN, MCI). Clear clustering is visible with distinct group boundaries. Same-class samples appear **densely grouped together**. Indicates the learned feature representations are **discriminative**. Suggests **strong separation** between categories in latent space. Overall, the visualization demonstrates meaningful internal feature organization.

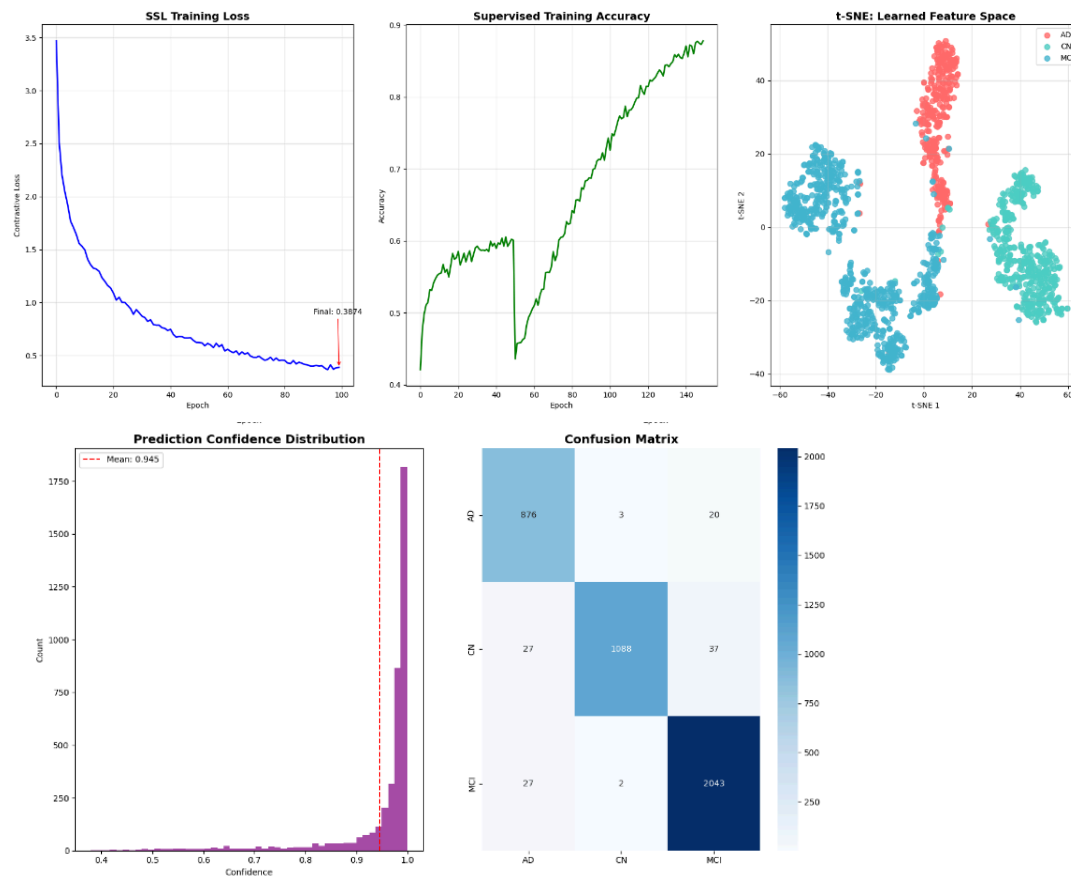
FINAL ACCURACY: 97.19%

Target: 98%+ | Achieved: 97.19%



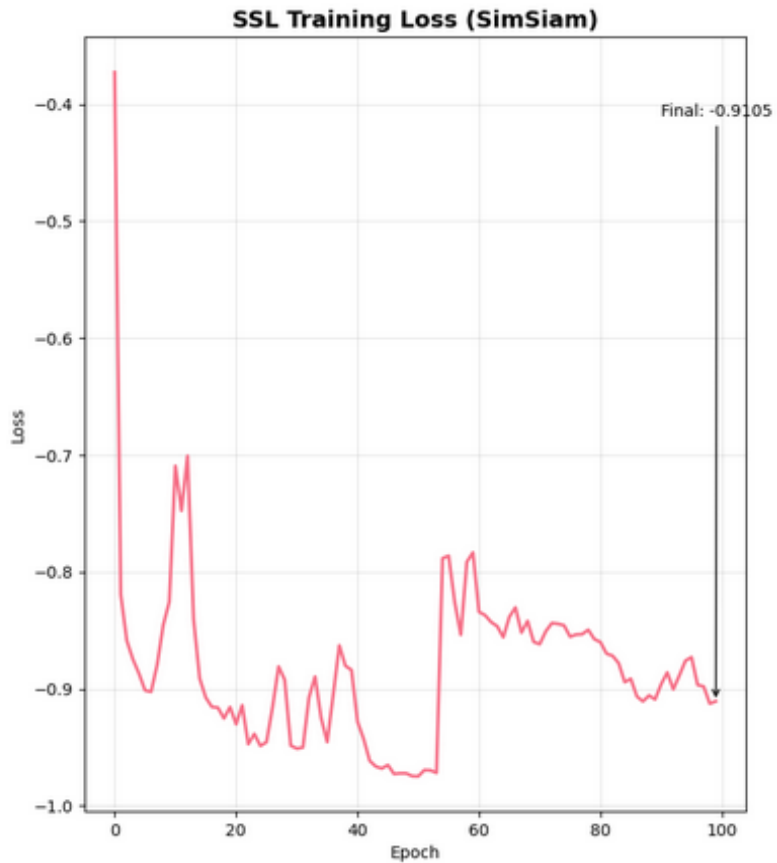
Close to target! Current: 97.19%





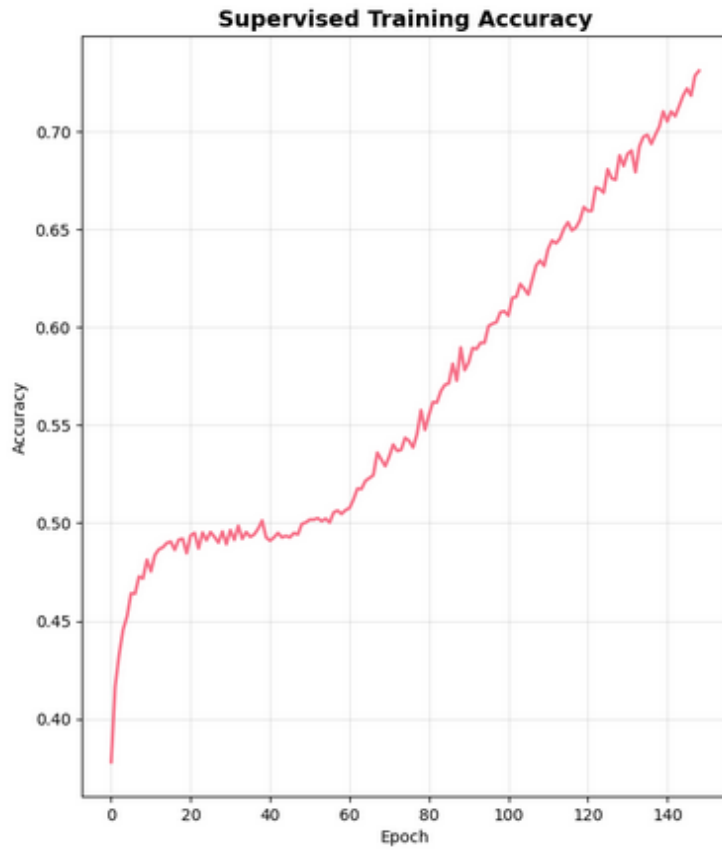
## SSL-3 SimSiam Method

**SSL Training Loss :**



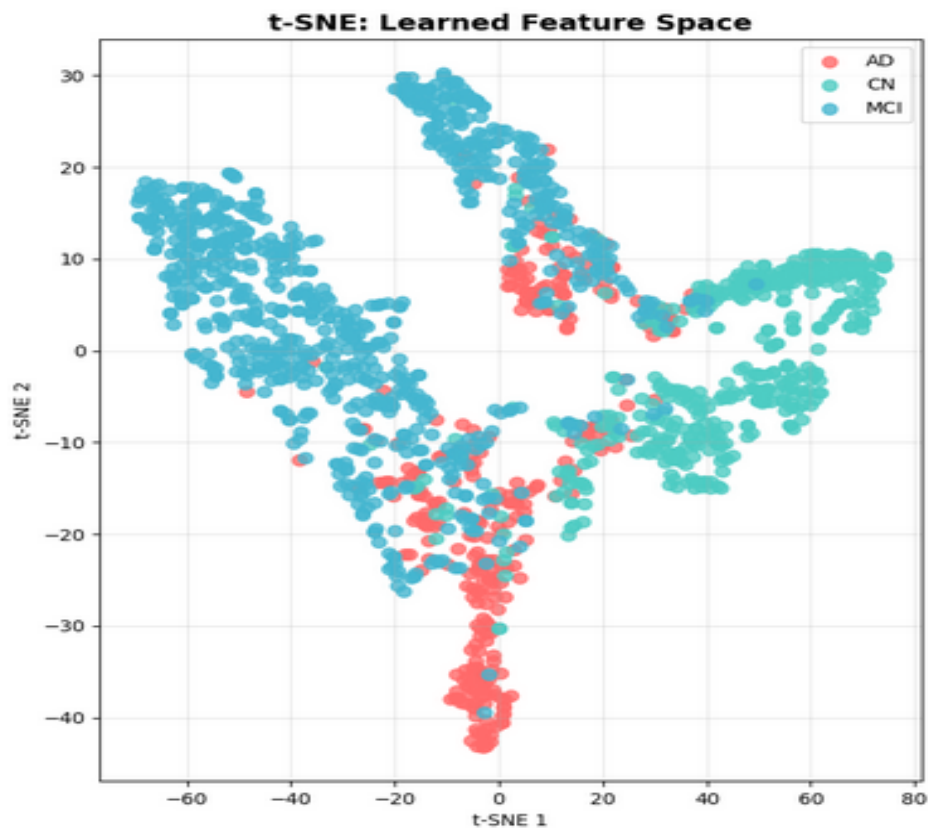
This graph shows a **SimSiam Training loss curve** labeled “Negative Start.” The y-axis represents **loss values** ranging from -1 to -0.4. The x-axis represents **epochs** from 0 to 100. This graph shows how the loss decreases during the training phase. Over the training epochs, the loss steadily goes down, indicating the model is learning meaningful representations. Near the **final epochs(100)**, the contrastive loss is at around **-0.9105**.

**Training Accuracy:**



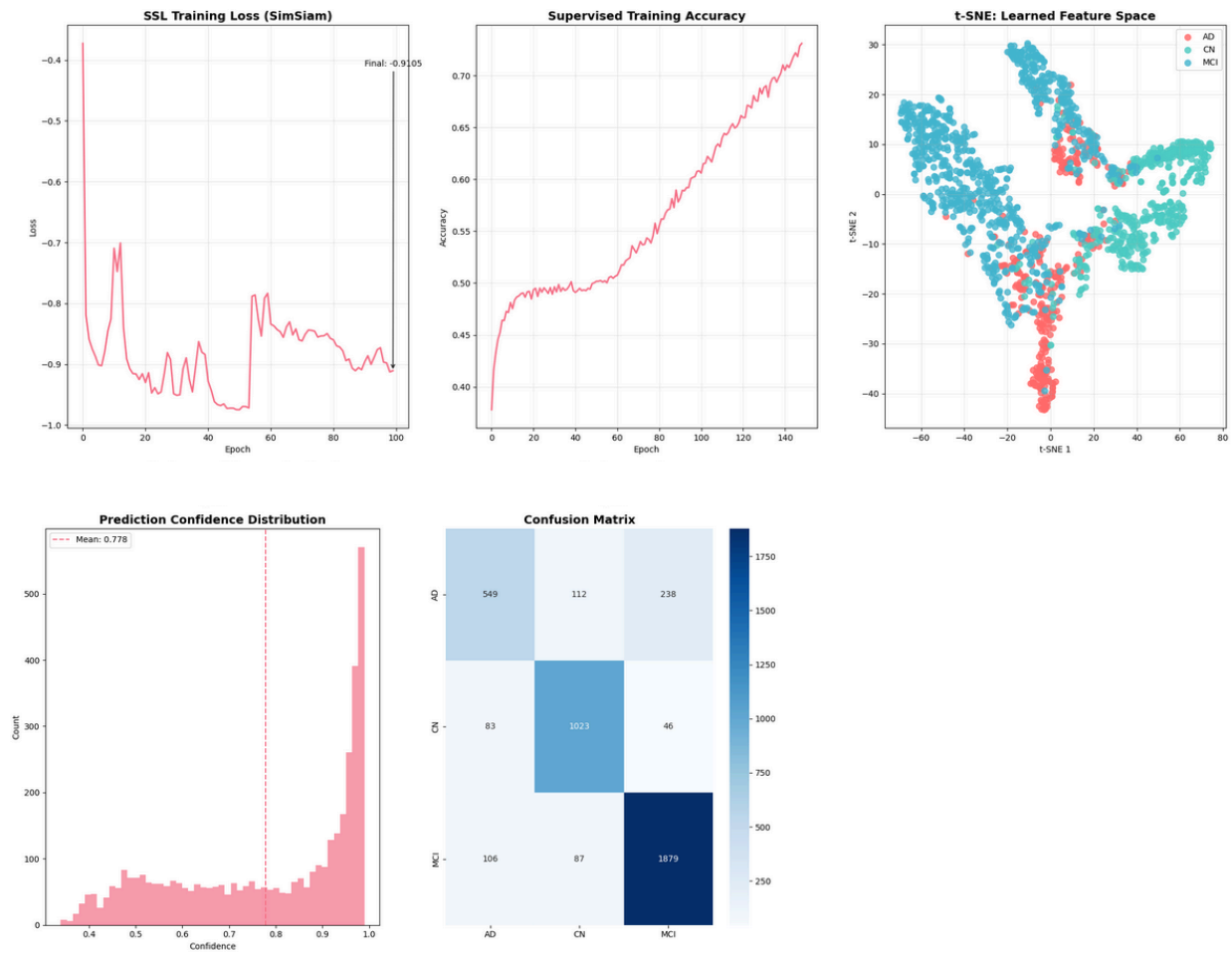
This graph displays the accuracy during supervised training. The y-axis represents **accuracy** ranging from **0.30 to 0.80**. The x-axis represents **epochs** from 0 to 140. At the beginning, accuracy increases sharply, then accuracy increases rapidly. Over time the improvements become smaller but continue to go up, then reaching **around ~75%**, which indicates Moderate performance on the labeled dataset.

**t-SNE Learned Feature Space:**



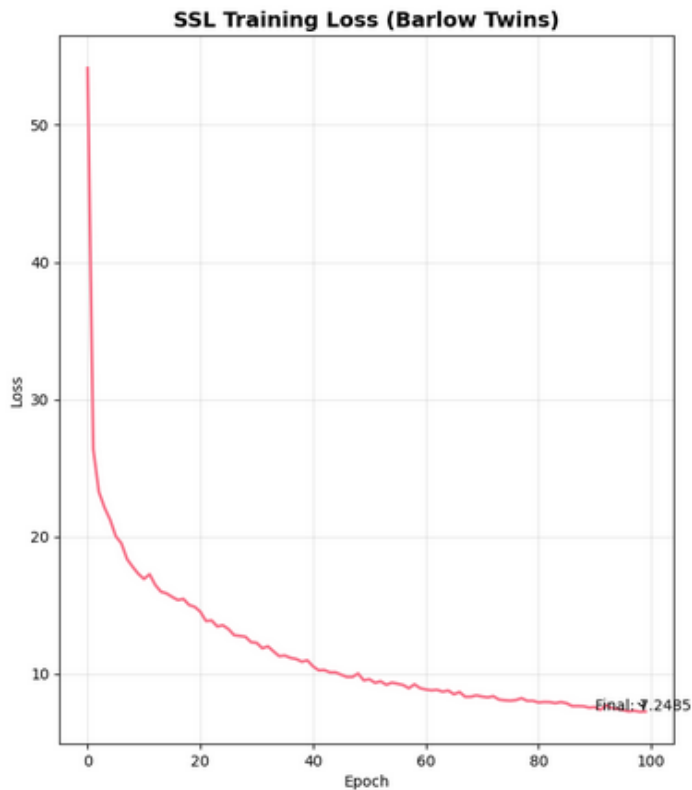
This shows the t-SNE projection of the learned feature space and also shows how the features learned by the SimSiam method are distributed in 2D space using t-SNE. Each data point represents a sample, and the different colors correspond to different classes such as AD, CN, and MCI. We can see that the clusters are somewhat formed, but there is still a sizable amount of mixing and overlap between classes. Some class members **appear spread out**, indicating that the learned representations are only moderately separable. Overall, SimSiam shows **partial clustering ability**, but the boundaries between different groups are **not fully separated**.

FINAL ACCURACY: 83.70%  
Target: 99% (depends on data, but SSL upgraded to a stronger SimSiam-style approach)



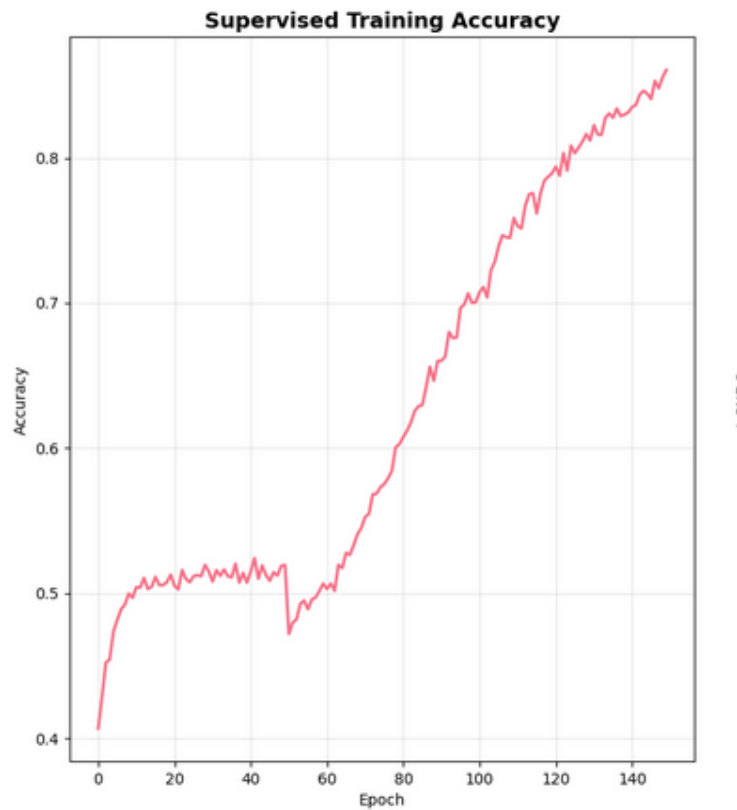
## SSL-4 Barlow Twins Method

## SSL Training Loss :



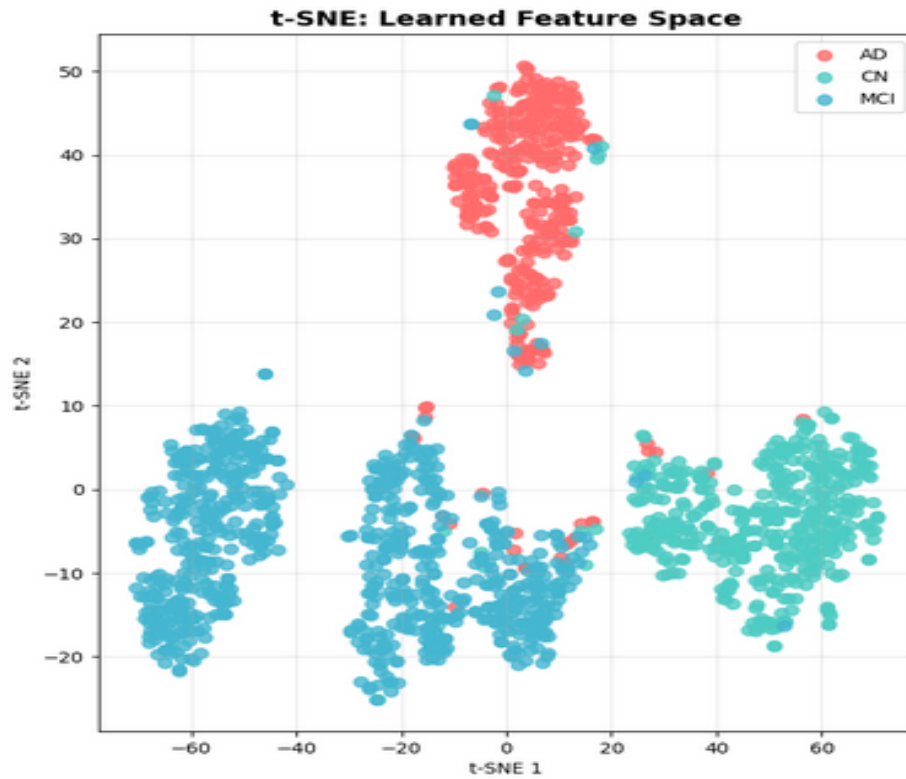
This graph shows a Barlow Twins training loss curve . The y-axis represents loss values ranging from **approximately 0 to 50 ups** .The x-axis shows training **epochs from 0 to 100**.The curve demonstrates a gradually decreasing loss throughout the SSL pretraining stage.The loss slope flattens toward the end, suggesting stable convergence and learned embedding quality.This improvement reflects successful unsupervised optimization of the feature encoder.The graph indicates that the model continuously reduces contrastive loss while learning data representations. Near the **final epochs(100)**, the contrastive loss at around **7.2485**.

## Training Accuracy:



This graph displays the accuracy during supervised training. The y-axis represents **accuracy** ranging from 0.4 to 0.9 .The x-axis represents **epochs** from 0 to 140. At the beginning, accuracy increases rapidly, but **around 45 epochs** it **decreases sharply**. Over time the improvements become smaller but continue to go up, then reaching **around ~85–90%**, which indicates strong performance on the labeled dataset.

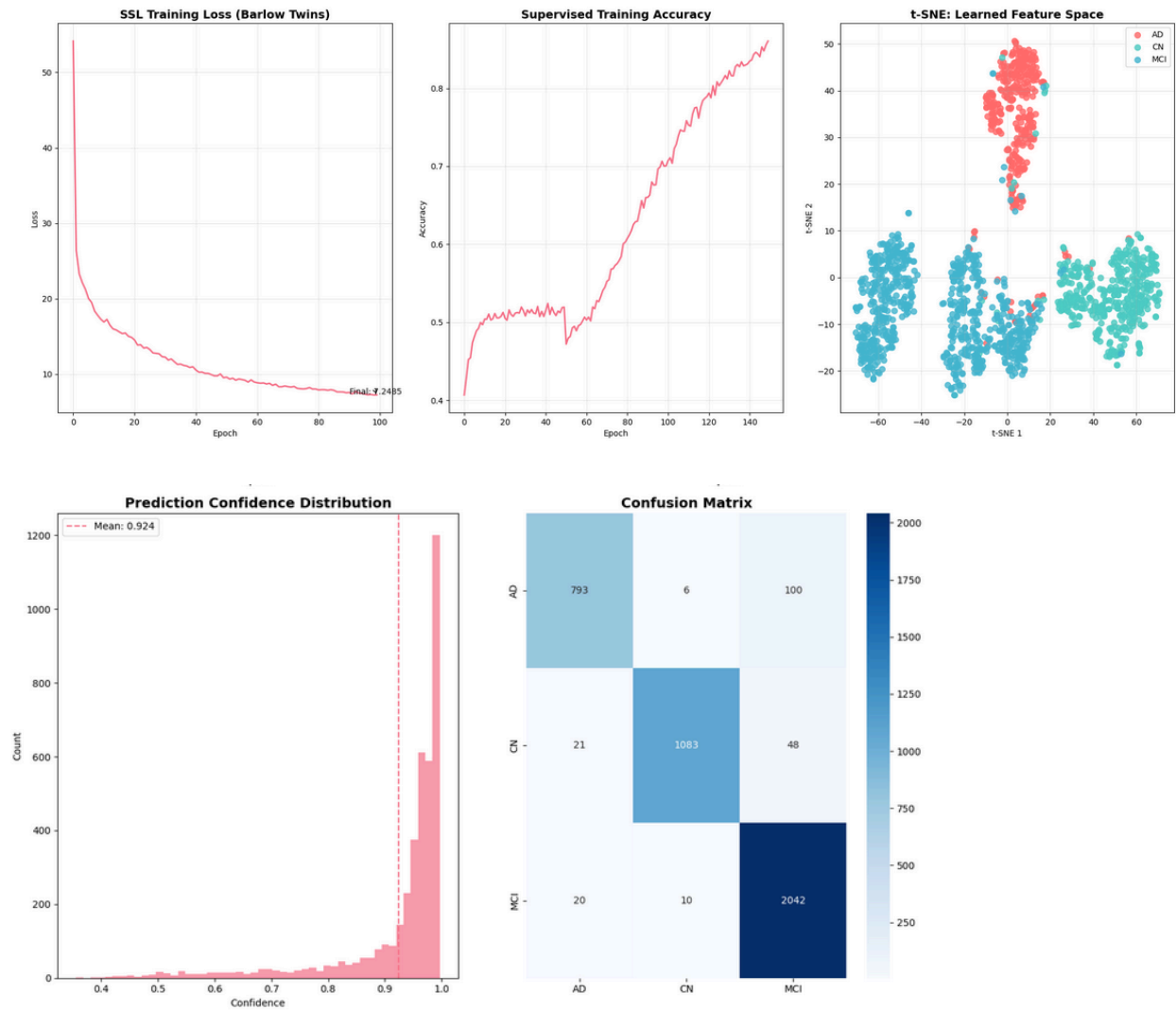
**t-SNE Learned Feature Space:**



This shows the t-SNE projection of the learned feature space and it shows the t-SNE projection of features learned by the **Barlow Twins** method. The feature clusters are much more clearly separated. Each class forms a separate group, and the amount of overlap is small. The feature representations appear cleaner and separate, meaning that the model is better at learning meaningful patterns from the data. As a result, Barlow Twins provide a **stronger separation** between the classes and is likely to **perform better** in downstream classification tasks.



FINAL TRAINING ACCURACY: 95.03%  
Target: 99% (actual result depends on data, but SSL upgraded to Barlow Twins)



Composition table :

| Method Name | SimCLR | SimSiam | Barlow Twins |
|-------------|--------|---------|--------------|
| Accuracy    | 97.17% | 83.90%  | 95.08%       |

